

Ontario Academy
5-2 Corby Rd. Markham, ON Canada
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Final Exam

Name: _____

Date: _____

MDM4U Data Management

Please read all the following instructions carefully before you start your exam.

- This is not an open book exam, other than a standard calculator or scrap paper, other aid, such as cheat sheets, a programmable calculator, electronic dictionaries, are not allowed.
- You are responsible to bring your own calculator. Sharing a calculator with someone else is strictly prohibited.
- Graphing calculators are NOT allowed for this test.
- Please show all your work on the space that provided in order to get full marks.

Good Luck!

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K/U	A	T	C
35	30	20	9

1. Identify the sampling method in each case, using descriptors such as simple random, systematic, stratified, multistage, voluntary-response, cluster, and convenience, and explain your choice. (8 Marks, K/U)

- a) You randomly select three schools. From each school, you select one class from each grade, then, you randomly select ten students from each class.
- b) The population of your town is 45% female and 55% male, so you randomly select 45 women and 55 men to survey.
- c) You walk into a mall and someone asks you to fill out a questionnaire.
- d) You write the names of all the students in your class on slips of paper, put the slips into a hat, and draw five of them.

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2. The scores on a test worth 40 marks are shown: (10 marks, App)

17 20 21 23 23 24 25 26 29 30 30 31 31 33 34 34 36 37 39 40

- a) Determine the mean, median, and mode of the scores
- b) Find the standard deviation
- c) Find the Inter-Quartier Range
- d) If a student scores at 80th percentile, what is the student's raw score?
- e) In which percentile are the students who scored 30 out of 40?
- f) If a student scores in the 70th percentile, what percent did the student get on the test?
- g) In which percentile is a student who gets 60% on the test?

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3. Identify the most likely type of causal relationship for each situation, assuming that the first variable is the independent variable; give explanation for your choice. (8marks, K/U)

- a) Toxic waste dumping increases in Lake Erie, and fish populations decline.
- b) You do well on tests when you wear a red shirt.
- c) When you eat a lot of chocolate, you get a lot of acne.
- d) As the number of police officers increases, so does the number of crimes.

4. Explain why outlier data point should be excluded from mean and standard deviation calculation. (3 marks, Comm)

5. How many arrangements are there of the letters for each word given?(4 marks, k/u)

a) arrangements

b) management

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6. Of the 140 students at Fairview High School, 52 signed up for French, 71 signed up for Spanish and 40 signed up for Japanese. The language students include 15 who are taking both French and Spanish, 8 who are taking Spanish and Japanese, 11 who are taking French and Japanese, and 2 who are taking all three language courses. (K/U)

a) Illustrate the enrolments with a Venn diagram (3 marks)

b) How many students are not taking any of these three language courses? (1 marks)

7. Simplify. (6 Marks, App)

a) $\frac{(2n)!}{(2n-2)!}$

b) $\frac{(n+2)!}{(n-2)!}$

8. Classify each pair of events as mutually exclusive or non-mutually exclusive. (5 marks, K/U)

	Event A	Event B	Mutually exclusive or not
a	Randomly drawing a grey sock from a drawer	Randomly drawing a wool sock from a drawer	
b	Having an even number of students in your class	Having an odd number of students in your class	
c	Calm weather at noon tomorrow	Stormy weather at noon tomorrow	
d	Sunny weather next week	Rainy weather next week	
e	Your birthday falling on a Saturday next year	Your birthday falling on a weekend next year	

9. Eleven children are to line up for a photograph (8marks, app)

- How many different arrangements are possible?
- How many arrangements are possible if Brenda is in the middle?
- How many arrangements are possible if Ahmed is on the far left and Yen is on the far right?
- How many arrangements are possible if Hanh and Brian must be together?

10. How many different sums of money can be made with six \$5 bill, three \$10 bills, and two \$50 bill? (4 marks, App)

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11. A experienced darts player has about a 15% chance of hitting triple-20 on any individual throw. (6 marks, App)

- a) Determine a probability distribution of missing the triple-20 for throwing up to six darts.
- b) What is the expected number of throws before there would be a miss?

12. You draw a card from a standard deck. What is the probability that the card is (4Marks, K/U)

- a) A jack or a heart?
- b) A face card or an ace?
- c) A spade or a heart?
- d) Which events in a) to c) are mutually exclusive events?

13. Most tests to detect diseases are not 100% accurate. Sometimes a test can give a false positive (indicating that you have the disease or condition when you do not) or a false negative (indicating that you do not have the disease or condition when you do). Suppose a test to check donated blood for the virus that causes chickenpox is found to give a positive result 98% of the time when the blood contains the virus and to give a false positive result 7% of the time when the virus is not present. If one pint of blood in 1000 pints contains the virus, find the probability that (12 marks, T/I)

- a) The blood has the virus and tests positive
- b) The blood does not have the virus and tests positive
- c) The blood does not have the virus and tests negative
- d) The blood has the virus and tests negative
- e) If the test is positive, it is correct
- f) If the test is negative, it is correct

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14. A bag contains six red marbles, five yellow marbles, and four black marbles. Marbles are selected from the bag one at a time without replacement until four marbles have been taken

a) Determine the probability distribution for choosing red marbles using (8 marks, T/I)

i) A tree diagram

ii) Combination

b) Create a probability histogram

c) Determine the expected number of red marbles

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15. Explain the differences and similarities between Binomial distribution, Geometric distribution, and hypergeometric distributions, given at least one example for each of those distributions. (6 marks, Comm)